

TAKADA, Japan

WMO: 476120

Lat: 37.1067N

Lon: 138.2467E

Elev: 15

StdP: 101.15

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	-1.7	-7.1	2.1	1.3	-6.1	2.3	1.6	8.6	4.0	7.7	3.3	2.6	180	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	33.6	25.6	32.3	25.3	31.1	24.8	26.4	31.6	25.9	30.7	25.3	29.8	3.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.2	29.0	24.5	19.5	28.5	24.0	18.9	28.0	82.4	31.8	80.0	30.9	77.7	29.9	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.4	5.4	DB	-5.0	36.6	1.2	1.3	-5.9	37.5	-6.6	38.3	-7.3	39.0	-8.2	40.0	(4)
(5)			WB	-5.7	27.5	1.1	0.6	-6.5	27.9	-7.2	28.3	-7.8	28.7	-8.6	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	2.6	2.8	6.0	11.7	17.1	21.1	25.5	26.9	22.7	16.8	10.7	5.4	(6)	
	DBStd	8.90	2.27	2.48	3.25	3.61	3.06	2.47	2.51	2.38	3.03	2.99	3.19	2.94	(7)	
	HDD10.0	763	231	203	132	22	0	0	0	0	0	1	29	144	(8)	
	HDD18.3	2270	489	436	383	199	60	5	0	0	2	67	230	399	(9)	
	CDD10.0	2285	0	1	7	74	221	333	481	523	382	211	50	3	(10)	
	CDD18.3	751	0	0	0	2	22	88	222	265	133	18	0	0	(11)	
	CDH23.3	6033	0	0	1	24	165	430	1849	2631	860	74	1	0	(12)	
	CDH26.7	2076	0	0	0	3	22	75	659	1054	252	11	0	0	(13)	
(14) Wind	WSAvg	2.3	2.7	2.5	2.5	2.6	2.4	2.0	1.8	1.9	1.9	2.1	2.3	2.6	(14)	
(15) Precipitation	PrecAvg	2517	367	230	174	99	87	137	193	159	191	192	287	403	(15)	
	PrecMax	3008	546	364	272	162	186	258	465	426	287	364	510	750	(16)	
	PrecMin	2011	192	83	96	20	39	52	27	24	47	110	111	166	(17)	
	PrecStd	292	91	71	44	35	39	56	105	88	68	59	89	130	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	15.0	21.2	26.0	28.8	30.8	34.6	36.0	33.4	28.0	22.2	17.1	(19)	
		MCWB	7.5	10.5	12.2	15.7	18.1	22.5	25.4	25.6	24.6	21.0	15.0	11.7	(20)	
	2%	DB	9.6	11.2	17.0	23.2	27.0	28.6	33.1	34.0	31.4	25.2	19.7	14.2	(21)	
		MCWB	5.7	6.7	9.7	14.3	17.6	21.6	25.5	25.8	24.3	19.0	14.1	9.6	(22)	
	5%	DB	7.8	9.1	14.4	20.7	25.2	27.2	31.8	32.9	29.6	23.5	17.9	12.1	(23)	
		MCWB	4.8	5.2	8.8	12.9	16.9	20.8	25.2	25.7	23.6	18.3	13.2	8.5	(24)	
	10%	DB	6.3	7.3	12.0	18.6	23.4	25.8	30.4	31.7	27.7	21.9	16.2	10.4	(25)	
		MCWB	3.8	4.3	7.7	12.2	16.4	20.3	24.6	25.4	22.6	17.2	12.5	7.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	10.0	12.8	17.1	20.5	24.4	27.1	27.4	26.1	22.1	16.7	12.6	(27)	
		MCDB	11.6	14.4	18.9	22.3	25.0	28.4	32.4	32.7	31.2	26.2	19.5	14.9	(28)	
	2%	WB	6.3	7.3	10.8	15.3	19.2	23.1	26.2	26.6	25.0	20.5	15.2	10.4	(29)	
		MCDB	8.7	10.1	15.5	20.9	24.1	26.2	31.3	31.9	29.7	23.5	18.0	13.3	(30)	
	5%	WB	5.2	5.8	9.4	14.1	18.2	22.2	25.6	26.1	24.2	19.3	14.2	8.9	(31)	
		MCDB	7.3	8.4	13.2	18.9	22.8	25.3	30.3	31.2	28.2	22.1	16.9	11.5	(32)	
	10%	WB	4.0	4.4	8.0	13.0	17.2	21.3	25.0	25.7	23.4	18.2	13.0	7.7	(33)	
		MCDB	6.1	7.0	11.9	17.4	21.8	24.6	29.3	30.3	26.8	20.9	15.6	10.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.8	8.7	10.4	10.1	8.1	7.3	7.9	7.9	8.6	8.4	6.7	(35)	
		MCDBR	8.9	10.7	13.8	14.5	13.3	10.3	9.6	9.7	9.8	10.2	11.2	10.0	(36)	
	5% WB	MCWBR	6.0	7.1	8.3	7.3	6.1	4.8	4.0	3.9	4.3	5.3	6.6	6.5	(37)	
		MCWBR	8.4	9.9	12.4	12.0	10.7	8.1	8.6	8.8	8.7	8.3	9.4	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.408	0.444	0.505	0.542	0.555	0.579	0.574	0.564	0.499	0.453	0.422	0.412	(40)	
		taud	1.970	1.886	1.826	1.839	1.872	1.887	1.961	1.991	2.123	2.148	2.142	2.037	(41)	
	Ebn at Noon	Ebn at Noon	731	758	753	754	752	732	732	729	755	750	722	697	(42)	
		Edh at Noon	136	168	196	204	201	198	182	174	144	128	115	120	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.72	2.52	3.63	4.66	5.33	5.05	4.79	4.92	3.88	3.04	2.07	1.54	(44)	
	RadStd	RadStd	0.14	0.23	0.29	0.39	0.55	0.36	0.71	0.51	0.30	0.26	0.17	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page